Section 6.1.2, where we aim to minimize the entropy in the *posterior distribution* of the quantity that we care about, in this case the location L.

How do we compute the entropy of the posterior distribution of L after inspecting a given location? This calculation requires us to reason about what we can conclude about L upon inspecting a given location, which is quite easy to do. Say we decide to inspect the first location x_1. According to our belief about L, there's 1 percent that L is this location, and 99 percent that L is one of the remaining slots.

- If L is indeed this location, then our posterior entropy about L drops to 0, as there's no more uncertainty about this quantity.
- Otherwise, the distribution of L gets updated to reflect this observation that z is not the first number of the list.

Figure 6.7 shows this process as a diagram, in which we need to update the distribution for L so that each of the 99 locations has 1 divided by 99, or roughly 1.01 percent, probability of containing z. Each of the 99 locations is still equally likely, but the probability of each location has gone up a bit, since we have ruled out the first location in this hypothetical scenario.

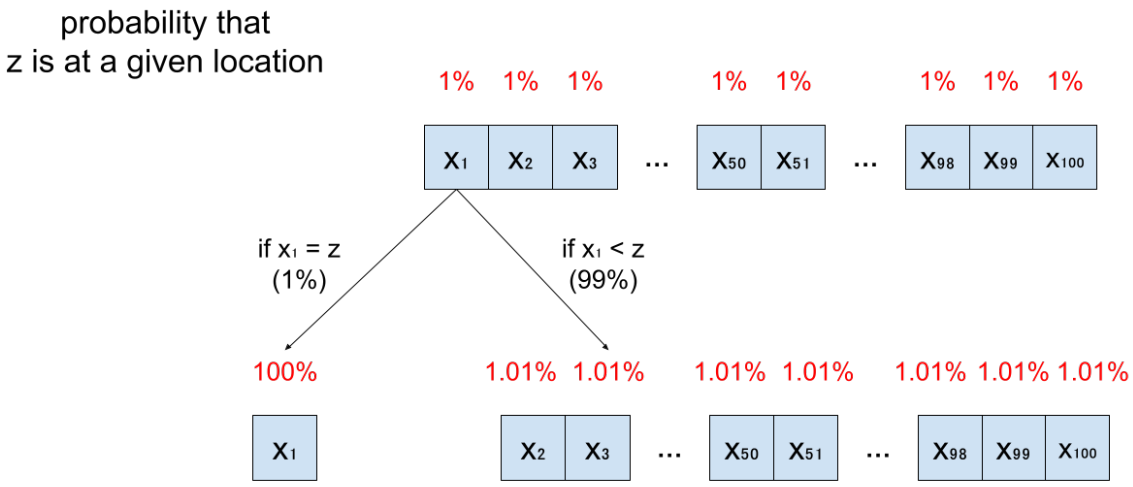


Figure 6.7 Posterior distributions of the location of target z within the 100-element list upon inspecting the first element. In each scenario, the probability that z is at a given location is updated accordingly.

NOTE Again, we're only considering the case where z exists in the list, so either the smallest element in the list x_1 is equal to z, or the former is less than the latter.

©Manning Publications Co. To comment go to liveBook
https://livebook.manning.com/#!/book/bayesian-optimization-in-action/discussion

Licensed to ashwin sampath <ashwin.manning@gmail.com>